



Vantage® Research Ultrasound System

Application Note - Vantage Example Scripts for Biomedical Applications
D00034 Rev. A

Summary

This documentation provides a matrix summary listing all available example scripts for Vantage™ 128 Systems and Vantage 256 Systems in the directory of Examples_Biomedical. In addition, a brief description of each example type that is associated with the first two columns of the matrix ("File Name" and "Description of Example") for Vantage 128 and 256 Systems, as well as Specialty_Applications, are provided. These notes are extracted from the documentation in the header in the example script source code.

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1. Imaging Mode Programming Example Scripts

The following scripts are based on the L11-5v probe example:

RyLns - Conventional Scanlines with Focused Beams

RyLns is a sequence programming file (for L11-5v Linear array) that uses 128 ray lines (for focused transmits) and receive acquisitions.

- Out of the 128 transmit available channels, only a certain number of transmit channels (denoted by numTx) are used, and numTx/2 transmitters are placed on each side of the center element when possible.
- All 128 receive channels are active for each acquisition.
- Usually, the number of ray lines is equal to (but not limited to) the number of transducer elements.

RyLnsMultiFocal - Scanlines with Multiple Focal Zones

RyLnsMultiFocal is a sequence programming file (for L11-5v Linear array) that uses 128 ray lines (for focused transmits) and 128 receive acquisitions.

- Of the 128 transmit channels with 3 different focal zones, the active transmit aperture is constrained based on user-entered transmit focus and f-number.
- All of the 128 receive channels are active for each acquisition.
- The receive acquisitions use 100% bandwidth to improve DMA transfers. Processing is asynchronous with respect to acquisition.

RyLnsSubFrmDma - Sub Frame DMA transfer

RyLnsSubFrmDma is a copy of the L11-5v_128RyLns script, modified to add a subframe to account for DMA transfers within the acquisition event sequence for each frame.

- This script displays CGD memory allocation at the MATLAB® command prompt when VSX loads and initializes the script.
- The value "numberOfAcquisitionBlocks" represents the amount of memory allocated within each channel group for the storage of receive data.
- One acquisition block is 128 samples per channel (and thus the size of an acquisition block is 8192 bytes = 128 samples * 2 bytes per sample * 32 channels); therefore, 128 * numberOfAcquisitionBlocks results in the total number of samples being allocated for each individual channel.

AcquireRF - RF Data Acquire Only (No Imaging)

AcquireRF is a sequence programming file (for L11-5v Linear array) used for acquiring RF data of a single plane wave transmit and receive acquisition.

- All 128 transmit and receive channels are active for each acquisition.
- External processing is used asynchronously with respect to acquisition.

Flash, Single Plane Wave

Flash is a sequence programming file (for L11-5v Linear array) used for a plane wave transmission and single receive acquisition.

- All 128 transmit and receive channels are active for each acquisition.
- Processing is asynchronous with respect to acquisition.

FlashAccum - Hardware Accumulate RF Data

FlashAccum is a sequence programming file (for L11-5v Linear array) used for plane wave transmit and receive by accumulating multiple acquisitions.

- All 128 transmit and receive channels are active for each acquisition. The processing of acquired data is asynchronous with respect to data acquisition process.

FlashAngles - Plane Waves with Multiple Steering Angles

FlashAngles is a sequence programming file (for L11-5v Linear array) used for plane wave transmits with multiple steering angles.

- All 128 transmit and receive channels are active for each acquisition.
- Processing is asynchronous with respect to acquisition.

FlashAnglesArbGP - ArbWave for Gaussian Pulse

FlashAnglesArbGP is a sequence programming file (for L11-5v Linear array) used for plane wave transmits of arbitrary waveforms with multiple steering angles.

- All 128 transmit and receive channels are active for each acquisition.
- Processing is asynchronous with respect to the acquisition.
- The Arbitrary Waveforms use a Gaussian Pulse (BW 60%) encoded (*per IUS Conference 2013 method - "Arbitrary Waveforms using a Tri-State Transmit Pulser", Flynn et al, IEEE/IUS Prague 2013*). The impulse response model used was computed for element 80 of an ATL L11-5v scanhead and applied to all element channels. The resulting encoding was tested on a different L11-5v scanhead and found to retain its effectiveness. Results on other L11-5v scanheads may vary.

FlashAnglesHI - Harmonic Imaging Using Pulse Inversion with Flash Angles

FlashAnglesHI is a sequence programming file (for L12-3v Linear array) used for pulse inversion (with flash angles) transmits and receive acquisitions.

FlashAnglesSC - Spatial Compounding (Incoherent Average)

FlashAnglesSC is a sequence programming file (for L11-5v Linear array) used for plane wave transmits with multiple steering angles and spatial compounding over 3-angle ranges.

- Acquisition is over "na" uniformly spaced angles over +/- 20 degrees. Reconstruction is on "na/3" angles at a time, with the detected result averaged multiplicatively.
- All 128 transmit and receive channels are active for each acquisition.
- Processing is asynchronous with respect to the acquisition.

FlashDoppler - Color Flow Doppler (CFD) and Power Doppler (PD)

FlashDoppler is a sequence programming file (for L11-5v Linear array) used for plane wave transmits for 2D (B-mode) and Doppler imaging.

- For the 2D (B-mode) imaging, "na" steered angles are used.
- For the Doppler ensemble, "ne" pulses are transmitted at a steering angle of dopAngle radians.
- This version uses TPC profiles to provide a separate high voltage level for Doppler.
- All 128 transmit and receive channels are active for each acquisition.
- Processing is asynchronous with respect to the acquisition.

FlashDopplerLF, Low-Flow (CFD & PD) with Long Ensembles

FlashDopplerLF is a sequence programming file (for L11-5v Linear array) used for plane wave transmits for 2D (B-mode) and Doppler imaging.

- For 2D (B-mode) imaging, na steered angles are used. 128 transmit and receive channels are active for each acquisition.
- For the Doppler ensemble, ne pulses with long ensemble lengths are transmitted at a steering angle of dopAngle radians. The preset of ensemble length is included via a switch statement on "EnsembleLength". 128 transmit and 64 receive channels are active for each Doppler ensemble.

- This version uses TPC profiles to provide a separate high voltage level for Doppler.
- Processing is asynchronous with respect to the acquisition.

FlashExtFunc - External Function

FlashExtFunc is a sequence programming file (for L11-5v Linear array) used to acquire RF data of single plane wave transmit and receive acquisition.

- All 128 transmit and receive channels are active for each acquisition.
- External processing is used synchronously with respect to the acquisition.

FlashExtTrig - External Triggering

FlashExtTrig is a sequence programming file (for L11-5v Linear array) used for plane wave transmit and single receive acquisition with an external trigger.

- All 128 transmit and receive channels are active for each acquisition.
- The external trigger-in and reconstruction/processing are synchronized with respect to the acquisition.
- To run this script with the VDAS hardware, users need to connect an external trigger source to the first trigger input.

FlashHFR - Very High Frame Rate Continuous Acquisition

File name: RunSetUpL11_5vFlashHFR_acquire.m - "High Frame Rate" acquisition for ultrafast imaging (script 1 of 2).

Note: This is a unique example that runs VSX automatically, and after closing the Control GUI, runs the post-processing script (script 2 of 2) automatically. Two scripts are a part of this example.

To support higher acquisition frame rates with reduced DMA overhead, this script acquires a large set of T/R acquisitions into each RcvBuffer 'super' frame, performing a transferToHost only after each group of "numAcqs" acquisitions. Even though each acquisition is intended to be reconstructed and processed into a display frame, live image reconstruction only processes the first acquisition in the super frame to not slow down real-time acquisition, while providing visual feedback to help the user collect the desired data. To post-process all acquisitions and superframes present in the RcvBuffer, the data is saved in a matfile when VSX quits (see the end of the script).

Note that the current system software does not permit running this same script to process all the acquisitions, even in playback (simulationMode=2), because a sequence can only reconstruct a maximum of 16 acquisitions in a given receive frame; as a result, the sequence cannot include all of the data transferred in one DMA (called a superframe).

A second script for reconstructing and displaying all acquisitions has been developed to be run in simulateMode=2 after this one, using the previously collected data. That script, SetUpL11_5vFlashHFR_reconAll.m, reconstructs and displays all acquisitions by defining ReconInfo structures for each acquisition and defining a Recon event for each display frame. Acquiring data using that script results in skipped superframes when processing cannot keep up with acquisition. To illustrate the HFR acquisition and display in simulation, the 'movePoints' function is invoked for every acquisition, resulting in a discontinuous apparent motion of the scatterers in "real-time". However, the motion is smooth on playback when post-processing each T/R acquisition with the "reconAll" script. For convenience, this script is currently set to launch VSX automatically. Upon quitting VSX (by closing the control GUI), the "reconAll" script is automatically invoked to process and display all the frames just collected.

Note: In simulateMode=1, this script runs once, fills all numFrames frames, and then freezes. In such a situation, close the GUI, which then launches the reconAll script automatically to demonstrate the post-processing of all acquisitions.

File name RunSetUpL11_5vFlashHFR_reconAll.m: "High Frame Rate" post-acquisition processing script (script 2 of 2):

To support higher acquisition frame rates with reduced DMA overhead, the first script (SetupL11_5vFlashHFR_acquire) acquires numAcqs acquisitions into each RcvBuffer "super" frame, performing a transferToHost after each numAcqs set. After quitting VSX, the RF data is saved in a matfile called 'RFdataHFR'. The real-time image reconstruction only processes the first subframe in the superframe to minimize processing time and yet provide real-time feedback during data collection. The data is saved in a matfile to permit the "clear all" at the beginning of every script. Note that in this HFR script, each acquisition leads to an image frame, and each DMA data transfer to the host contains one receive "super" frame.

This second script has been written to reconstruct and display all acquisitions. To be compatible with the acquired data, it must use the same RcvBuffer size and is intended to be run in simulateMode=2 using the previously collected data. It works in real-time as well; however, the processing cannot keep up with acquisition, and some superframes are skipped but the script displays all sub-frames within a superframe.

To illustrate the HFR effect in simulation, the 'movePoints' function is invoked (in SetupL11_5vFlashHFR_acquire) for every acquisition, resulting in a discontinuous apparent motion of the scatterers in "real-time" when processing only one acquisition per superframe. However, the motion is smooth on playback when processing each T/R acquisition.

FlashSL - Steered Linear Probe

FlashSL is a sequence programming file (for L11-5v Linear array) used for a plane wave transmit with steered linear array and single receive acquisition.

- All 128 transmit and receive channels are active for each acquisition.
- Processing is asynchronous with respect to acquisition.

FlashSync - Synchronous Scripting

FlashSync is a sequence programming file (for L11-5v Linear array) used for a plane wave transmit and single receive acquisition synchronized to software reconstruction/processing sequences.

- All 128 transmit and receive channels are active for each acquisition.

Ideal - Full Matrix Capture

This is a sequence programming file (for L11-5v Linear array) used for ideal synthetic aperture reconstruction.

- For each acquisition, one channel is active for transmit and 128 channels are active for receive.
- Reconstruction and processing are synchronous with respect to acquisition.

SingleBeam - One Focused Beam

SingleBeam is a sequence programming file (for L11-5v Linear array) used for single beam focused transmit.

- 96 transmit and 128 receive channels are active for each acquisition.
- Processing are synchronous with respect to acquisition.

SpectralDoppler - Spectral Doppler with Audio

SpectralDoppler is a sequence programming file (for L11-5v Linear array) used for plane wave transmits for 2D and Doppler imaging.

- For the 2D image, a single plane wave transmit is used. All transmit and receive channels are active for each acquisition.
- For the Doppler ensemble, multiple pulses are transmitted at a steering angle of dopAngle radians. 128 transmit and 64 receive channels are active for each acquisition. The Doppler measurement region is the region under the central 64 transducer elements.
- The processing is asynchronous with respect to acquisition.
- An acquisition 'frame' consists of the number of transmit/receive events that equals 10 msec of time. Doppler and 2D acquisitions are interleaved.
- Doppler PRF is quantized such that the acquisition PRF (twice the Doppler PRF) has a PRI of integer microseconds.
- For the Doppler measurement, the sample volume in the measurement region is selected by clicking the left mouse button in the DisplayWindow. The sample volume is selected as a single I, Q pixel whose location is indicated by a red circle. This action also switches the processing mode to Doppler only. Clicking the right mouse button cancels the Doppler processing and returns to 2D-only processing. While the processing operates in duplex mode - either 2D or Doppler, the acquisition is always the same, allowing Doppler operation in RcvData Loop playback mode.

TripleBeam - Three Focused Beams

TripleBeam is a sequence programming file (for L11-5v Linear array) used for 3 focused beams on transmit.

- For each acquisition, 96 channels are active for transmit and 128 channels are active for receive.
- Processing is asynchronous with respect to acquisition.

WideBeam - Weakly Focused Overlapping Beams

WideBeam generates a .mat Sequence Object file for L11_5v Linear array wide beam transmit.

- The transmit aperture size can be set by the variable - numTX. The transmit aperture is translated across the array with the number of rays set by the variable - numRays.
- The receive aperture always covers the full 128-element aperture.

WideBeamHI - Harmonic Imaging with Pulse Inversion Using WideBeam

WideBeamHI is a sequence programming file (for L11-5v Linear array) used for widebeam pulse inversion transmits and receive acquisitions.

- The transmit aperture size is set by the variable - numTX, and the number of pulse-inversion transmits are set by the variable - numRays.
- All 128 receive channels are active for each acquisition.
- Processing is asynchronous with respect to the acquisition.

2. 67% Bandwidth and Interleave for High-Frequency Probes

128RyLns67BW - Scanline - 67% BandWidth

128RyLns67BW is a sequence programming file (for L22-14v Linear array) used for 128 ray lines (focus transmits) and 128 receive acquisitions.

- Of the 128 transmit channels, the active transmit aperture is limited based on user-entered transmit focus depth and f-number.
- All 128 receive channels are active for each acquisition.
- This script uses 4/3 sampling with A/D sample rate of 50 MHz followed by a high-pass filter and subsampling by 2, to produce the 25 MHz output sample rate with 18.75 MHz processing center frequency. Transmit is at 17.8 MHz.

- Processing is asynchronous with respect to the acquisition.

Flash67BW - Plane Wave - 67% BandWidth

Flash67BW is a sequence programming file (for L22-14v Linear array) used for plane wave transmit and single receive acquisitions.

- All 128 transmit and receive channels are active for each acquisition.
- This script uses 4/3 sampling with an A/D sample rate of 50 MHz followed by a high-pass filter and subsampling by 2, to produce the 25 MHz output sample rate with 18.75 MHz processing center frequency. Transmit is at 17.8 MHz.
- Processing is asynchronous with respect to the acquisition.

FlashAngles67BW - Multiple Plane Waves - 67% BandWidth

FlashAngles67BW is a sequence programming file (for L22-14v Linear array) used for a plane wave transmit with multiple steering angles.

- All 128 transmit and receive channels are active for each acquisition.
- This script uses 4/3 sampling with an A/D sample rate of 50 MHz followed by a high-pass filter and subsampling by 2, to produce the 25 MHz output sample rate with 18.75 MHz processing center frequency. Transmit is at 17.8 MHz.
- Processing is asynchronous with respect to the acquisition.

FlashAnglesInterleave - 2x Interleaved Sampling (125 MHz)

FlashAnglesInterleave is a sequence programming file (for L22-14v Linear array) used for plane wave transmits with multiple steering angles using 4X sampling.

- All 128 transmit and receive channels are active for each acquisition.
- This script demonstrates the "interleaved sampling" mechanism for RF data acquisition.
- In this script, the L22-14v is operated at a nominal center frequency for Recon processing of 20.83 MHz, with 4X sampling so the required RF data acquisition sample rate is 83.3 MHz. Since the Vantage™ Hardware System cannot acquire data at sample rates above 62.5 MHz, we acquire two sets of receive data for each line, at an A/D sample rate of 41.7 MHz.
- These pairs of acquisition data lines are then interleaved in an external processing function, to produce a single composite line sampled at 83.3 MHz as expected by Recon. The data acquired in the second acquisition of each pair must have its data shifted by half a sample period, so it can be interleaved with the first line. This is accomplished by adding an offset to the transmit delays for the second line equal to that half-sample interval; this in effect shifts each receive sample's data earlier by the desired one-half-sample period (other than the interleaved acquisition at 83.3 MHz, this script is identical to L22-14vFlashAngles which uses acquisition at 62.5 MHz with no interleave).

WideBeam Spatial Compounding Interleave - 2x Interleaved Sampling (125 MHz)

This sequence programming file is for the L35-16v Linear array wide beam scan, with 3 steering angles for spatial compounding using 4X interleaved sampling.

- Acquisitions for each steering angle are acquired into separate RcvBuffers, and the most recent frame from each buffer is reconstructed. A sliding 3-frame averaging window is used on the processed image frames to combine the images.
- This script uses the "interleaved sampling" mechanism for RF data acquisition.
- The L35-16v is operated at a nominal center frequency for Recon processing of 31.25 MHz, with 4X sampling so the required RF data acquisition sample rate is 125 MHz. Since the Vantage Hardware System cannot acquire data at sample rates above 62.5 MHz, we acquire two sets of receive data for each line, at an A/D sample rate of 62.5 MHz. These pairs of acquisition data lines are then

interleaved in a 'Pre' routine for the reconstruction processing to produce a single composite line sampled at 125 MHz.

- The data acquired in the second acquisition of each pair must have its data shifted by half a sample period, so it can be interleaved with the first line. This is accomplished by adding an offset to the transmit delays for the first acquisition equal to that half-sample interval; this in effect shifts each receive sample's data earlier by the desired one-half-sample period.

3. High Image Quality scripts, based on L11-5v probe examples

WideBeamDoppler with Enhanced GUI for Clinical Applications

Based on the FlashDoppler example, FlashDopplerC provides greater functionality, which includes the following:

- A slider for PRF (velocity range) adjustment
- An outline on the display showing the area undergoing Doppler processing
- A button group to turn on/off the Doppler display
- A slider for image depth adjustment
- A slider for Doppler angle adjustment

WideBeamSC and WideBeam with Spatial Compounding

This generates a sequence program for the L11_5v linear array using wide beam transmit scanning with spatial compounding.

- Three wide beam scans are performed using P.numRays scan lines in each frame, with different steering directions (steered left, no steering, steered right).
- The frames are averaged after image processing using a running average that updates each frame. The transmit aperture size can be set by the variable - numTX. The receive aperture always covers the full 128-element aperture.
- The PData regions are generated to provide an overlap of 3 regions at every pixel.
- The transmit focus is set to below the bottom of the PData region to generate a wide beam that is as wide at the bottom of a single region.

WideBeamHISC and WideBeamSC with Harmonic Imaging

This generates a sequence program for the L11_5v linear array using wide beam transmit scanning with harmonic imaging with spatial compounding.

- Three wide beam scans are performed with P.numRays scan lines in each frame using pulse inversion averaging for each beam direction. Each scan is made with different steering directions (steered left, no steering, steered right).
- The frames are averaged after image processing using a running average that updates each frame. The transmit aperture size can be set by the variable - numTX. The receive aperture always covers the full 128-element aperture. This version acquires all three steering directions into a 'super' frame to allow asynchronous acquisition and processing.
- The PData regions are generated to provide an overlap of 3 regions at every pixel.
- The transmit focus is set to below the bottom of the PData region to generate a wide beam that is as wide at the bottom of a single region.

4. Special Application Examples

ColorDopplerImagingExternal - External Functions for Doppler (in MATLAB)

Color Doppler Imaging by external MATLAB function processing performs WF, correlation/est, detection, and AGC.

This generates a .mat Sequence Object file for flash transmit Doppler imaging.

- For the 2D image, no flat focus transmits at steered angles are used.
- For the Doppler ensemble, no flat focus pulses are transmitted at a steering angle of dopAngle radians, receiving on the central 64 elements.

Customized2DArray, Circular Aperture, 2D Planar Array 3D Imaging

This sets up the file for Conical scan format, flash transmit.

- Transducer has 256 elements arranged in a 2D grid (16x16).
- The angle of the scan is alpha, which is nominally around 20 degrees.
- The 3D volume region is a cube with the transducer centered in one face. The region imaged is an expanding cone, from a virtual apex behind the array, as shown in the script.

DualTransducer

File name: SetUpDualXdcr.m

This is an example script to illustrate the use of two separate transducers imaging simultaneously on a Vantage 256 system. For this example, an L7-4 transducer is connected to the left connector and a P6-3 transducer is connected to the right connector. This script is a combination of the "L7-4Flash" and the "P6-3Flash" example scripts, with both of them running in parallel but independently of each other in the same script. To accomplish this, we define separate structures for PData, Recon, and ImageDisplay for each transducer. The script requires a single Trans structure; however, it is defined as if it were a 256-element transducer using both connectors. Within that shared structure elements, elements 1:128 correspond to the L7-4 transducer, and elements 129:256 correspond to the P6-3 transducer. The TX and Receive.Apod arrays for each transducer have zeros in the channel positions representing the other transducer. Note that the GUI controls modify settings for the L7-4 transducer only, since it is indexed first in the relevant structures. Additional GUI control object(s) can be created to switch between probes. Two separate TPC profiles are used, to allow independent control of the transmit voltage for each transducer using P1 and P2 HV sliders.

GPUFilterRF

File name: SetUp_FilterRFGPU.m

This sequence programming file is for a Linear array used for acquiring RF data of a single plane wave transmit and receive acquisition.

- All 128 transmit and receive channels are active for each acquisition.
- External processing is used synchronously with respect to acquisition.
- This script demonstrates GPU usage with the Verasonics system, where the GPU functionality is accessed through the MathWorks Parallel Computing Toolbox. The example executes filtering on acquired RF data prior to reconstruction. The acquisition is a single planewave per image. The transducer used is L11-5v.
- Three RF frames are employed, and the filtering is managed using the External Process functions filterChannelRFGPU.m.
- The filter example chosen is an FIR bandpass centered at the transducer frequency. To adjust the length of the filter, modify the setup script variable filterCoeffLength.

FUS2D

FUS2D is a licensed feature for the customer who purchased the FUS Elite System, which is the combination of a HIFU transducer, an imaging array, and the HIFU configuration. A graphical user interface is provided to help the user design the treatment plan. Besides the GUI, two SetUp scripts are provided.

File name: SetUpFUS2D.m

This script is provided for the user to simply test the beam steering feature. Currently, only one focus is predefined and different UI controls are generated based on the pairing number. If the HIFU array is annular, i.e., FUS Transducer Packages -01 through -03, only one slider with the Z direction will appear. For the FUS Transducer Packages -04 through -06, three sliders with Z, R, and Rotation Angle are generated for focal position adjustment.

The three sliders are defined as follows:

- Z is the axial location of the focus. The default value, 0, is the natural focus. Positive values move the focus deeper and negative values move the focus toward the transducer.
- R is the lateral distance between the focus and the axial axis.
- Rotation Angle is the angle between the focus and the Bmode plane, which is also the X-Z plane along element 1 of the HIFU array.

The user can adjust the slider and voltage to have a better focal display on the film target. If the user needs to use multiple foci in the SetUp script, the callback functions need to be modified accordingly.

File name: SetUpHIFUSystem.m

This file shows how to correctly program the HIFU system in the setup script. Compared to the regular imaging script, comment lines using the prefix HIFU, identify the changes added to allow the script to exercise a TPC Profile 5 HIFU transmit-only event. No image is created.

The HIFU transducer should be connected to connector 2 and this script tests the HIFU transducer only, regardless of the connection of the imaging array on connector 1. If the HIFU transducer is connected, ensure that the HIFU transducer is coupled with water.

Note: Do NOT expose the HIFU transducer to the air while running VSX.

Low-Frequency Option

Since the Vantage Software release 3.1, Verasonics supports a low-frequency configuration. A few examples are provided to exercise transmit and receive at low frequencies.

Microvascular

This script allows users to capture reconstructed ensembles of IQ data appropriate for microvascular processing. An auxiliary Doppler sequence (the “audit” sequence) is processed and displayed in real-time so the user may locate the desired scanning region to be captured.

MultiSystemClockSynchronization

Examples are provided to synchronize two or four Vantage Systems.

- BNC cables can be used for two systems. For more details, refer to the application note, PA_Synch_Two Vantage Systems.
- The SyncModule can be used for 2 or 4 systems. For more details, refer to the application note, PA_Synch_Multiple_Vantage_Systems.
- In addition, for example scripts ending with “TCPIP”, use TCPIP communication to achieve real-time data transfer.

NonlinearImaging_PulseInversion

File name: SetUpL11_5vFlashAngles_PulseInv.m

This script provides an example of Pulse Inversion (PI) imaging using multiple-angle unfocused waves, for contrast agent imaging or tissue harmonic imaging.

- A broadband probe is required; this script is for the L11-5v transducer.
- Tissue nonlinearity is detectable but often too weak to be practical with unfocussed transmits; beam focusing greatly improves PI results in tissue.

- The 64-channel synthetic aperture example script is modified to do two transmits per pulse inversion pair, using all 128 channels. Each acquisition is transferred to the host for examination, and this approach easily permits adjusting the relative contribution of each pulse to explore the relative weighting of the fundamental and the second harmonic images. Reconstruction processing is faster if the summation is done in hardware accumulation before DMA; in such an implementation, the Receive gain can be used to adjust the relative weighting between individual and summed results.
- An external function provides plotting of 3 RF data channels, for the first acquisition and the sum of the 2 inverse pulses. While this slows down the frame rate due to the time required for MATLAB plotting, it is easy to comment out the external processing event for faster frame rates. Refer to the function "plotThreeChannels(RData)" for which channels to plot.
- This example also illustrates how to make a Setup script compatible with both Vantage and V1, using 2 helper functions that convert Vantage TW specifications into V1 TW commands, for the 'parametric' type. (Refer to Misc. folder in ExampleScripts)
- This example also uses the nonlinear acoustic simulation capability added in V2.9.0, which permits the use of the simulator to help develop a nonlinear imaging script. This example uses cells (with the %% delimiter) and runs VSX automatically after defining the variable "filename" as a string with the name of the matfile created by running this script.

PhotoAcoustics - Triggering a Laser using 1-Way Reconstruction

PhotoAcoustic application example script:

- This generates a .mat Sequence Object file for FlashAngles B-mode and PhotoAcoustic (PA) imaging with the L11-5v transducer and runs VSX automatically.
- It is modified from the SetupL11-5vFlashDoppler script and retains some extra flexibility that is not required for PA imaging. For example, PData, P, and other structures can be identical for both modes but are separate here for added flexibility.
- The PA data is displayed using the split palette colormap overlay method, using the Power Doppler colormap. Note that the compression slider does not work properly when using split palettes. When running in simulation (with oneway=0), the media points are imaged as though the medium moves between B-mode and PA acquisitions, and thus are slightly offset from each other to better visualize each result. The B-mode or PA mode images can be set to dominate the display by adjusting gain and compression and color priority controls.
- For the 2D (B-mode) image, 'na' flat focus transmits at steered angles are used.
- For the PA acquisition, 'ne' T/R events with output trigger for an external source (laser) are performed, and use receive-only reconstruction (when oneway=1).
- The sequence waits for an input trigger (e.g. from a laser flash lamp), then pauses while the laser pumps (flash2Qdelay microsecs), and finally fires the Q-switch using a trigger out that also begins receiving acquisition. This approach results in a very low jitter between the laser pulse and the ultrasound.
- Multiple PA acquisitions are reconstructed and then coherently averaged by accumulation in the Inter Buffer. Incoherent averaging is possible by summing in the Image buffer instead and relaxes jitter requirements. Though not implemented here, the RF data can be accumulated in hardware prior to transfer using the Receive.mode attribute; doing so also requires precise synchronization with the laser.
- The receive-only reconstruction mode is not currently supported in simulation, so to test the PA code we simulate by turning on the transmitters for a conventional T/R acquisition and

reconstruction. Toggling between modes is easily done using the 'oneway' parameter (=0 turn on transmitters, =1 turn off transmitters for real PA acquisition) below.

- Two TGC structures are defined, but only TGC(1) is set up to be controlled by the GUI sliders. To use the TGC slider controls for the PA acquisitions, use TGC(1) for PA and TGC(2) for 2D.

RadiationForce_ShearWaveVisualization, Extended Transmit “push”

File name SetupL7_4ShearWaveImaging.m:

This generates a .mat Sequence Object file for L7-4 Linear flash transmit with Shearwave visualization. An example of Changing startevent on conditional coding is provided. Refer to "ShearWaveImaging_Documentation" for further documentation.

Probe Temperature Sensing

File name SetupL11_5ThermistorDemo.m:

The Vantage System provides hardware circuitry and software utilities to allow monitoring of one or two transducer temperature sense thermistors connected through dedicated pins at the probe connector, for probes that provide this feature. The system can be programmed to automatically monitor the temperature sensors while a script is running and exit the script with an error condition if a predetermined temperature limit is exceeded. The temperatures can also be read from the hardware system whenever desired, for display or other purposes. At the hardware level, the system simply measures the resistance of the device connected to the thermistor signals at the probe connector. If that device is a thermistor, you can translate the measured resistance into a temperature, but you can also use this feature to measure the resistance of some other type of sensor or control within the probe. An AppNote is also provided as a reference.

Recon All IQData

File name SetupL11_5vFlashAngles_IQall.m:

Usually only one InterBuffer frame, i.e. IQData, is used in our example script because only the latest frame is reconstructed. If the user would like to save the beam-formed data (IQData) of each frame, the setting of Recon/ReconInfo must be correct. This example illustrates how to have the correct Recon/ReconInfo setting for IQ data of each angle and each frame.

SaveRF

File name: SetupL11_5Flash_saveRFbutton.m

This example is of imaging with single plane wave transmit with a push button control to save RcvData in the workspace. This example script shows the following:

- How to add a push button using UI.Control structure
- How to save RF data after freezing data acquisition. The RcvData will not be copied to the MATLAB workspace until the freeze operation is performed.

The saved RcvData can be reprocessed using simulation mode 2 because the file size is not reduced by eliminating all zeros. The Resource.RcvBuffer.LastFrame is also saved for reference.

File name: SetupL11_5FlashAngles_saveRFallFrames.m

This is an example of plane wave imaging with steering angle transmits and saving RF data in real-time. This script shows how to use an external function to save the RF data of all frames in a receive buffer in real-time. In this case, only non-zeros channel data is saved to reduce the saving time. The default frame limit is set to 100 RcvBuffers. You can change to a bigger number for more buffers.

File name: RunSetupL11_5FlashHFR_saveRFperFrame.m

This is an example of the "High Frame Rate" acquisition for ultrafast imaging with saving RF data per frame in real-time. This script shows how to use an external function to save the RF data of each frame

in real-time. In this case, only non-zero channel data is saved to reduce the saving time. The default frame limit is set to 100 "superframes". You can change to a bigger number for more superframes.

	System	Vantage 512		Vantage 256 / 128 / 64 LE																					Vantage 64 / 32 LE										
File Name (root)	Description of Example	UTA-512-Direct	UTA-2048-MUX	UTA-260-S and 260-D											UTA-408-GE													UTA-156-U		UTA-1024-MUX [2]	UTA-260-MUX				
Setup_Probe_root.m		L512tst	L512tst	P4-2v	C5-2v	L11-4v	L11-5v	L12-3v	L22-8v	L22-14v/vXL35-16v	L38-22v	IP-104	IP-105	L11-5gH	P4-2gH	C5-2gH	4CD	9LD	C1-6D	IC5-9D	L3-12D	M5ScDRC6gV	RC15gV	L39-21gD	[4]	L14-5u	SA4-2-24u	3 MHz	P4-2v	C5-2v	L11-4v	L11-5v	L22-14v/LF	[3]	
RunFlashHFR	Very High Frame Rate Continuous Acquisition					★L	★L							★L																					
###RyLns	Conventional Scanlines with Focused Beams	★		★L	★L	★L	★L	★	★	★L		★	★	★	★L	★L	★L								★						★L	★L	★L	★L	★L
###RyLnsMultiFocal	Scanlines with Multiple Focal Zones					★L	★L	★						★L																					
AcquireRF	RF Data Acquire only — no imaging					★L	★L							★L																					
Flash	Single Plane Wave Imaging	★	★	★L	★L	★L	★L	★L		★		★	★	★	★L	★L	★L	★	★	★	★	★		★	★				★		★L	★L	★L	★L	★L
FlashAngles	Multiple angle plane waves/ Coherent compounding		★	★L	★L	★L	★L	★L	★	★				★L	★L	★L	★	★	★	★	★		★	★				★		★L	★L	★L	★L	★L	
FlashAnglesArbGP	Gaussian Pulse shape using AWG					★	★							★																					
FlashAnglesHI	Harmonic Imaging using Pulse Inversion							★																											
FlashAnglesSC	Spatial Compounding/ Incoherent Average					★L	★L							★L																					
FlashDoppler	Color Flow Doppler (CFD), Power Doppler (PD)			★L	★L	★L	★L	★L	★	★	★			★L	★L	★L		★									★	★			★L	★	★L	★L	★L
FlashDopplerLF	Low-Flow (CFD & PD) with Long Ensembles				★L	★L	★L			★				★L		★L																			
FlashSL	Steered Linear Transmit Beams					★L	★L							★L																					
Ideal	Full Matrix Capture (FMC)/ Synthetic Aperture					★L	★L							★L									★	★											
SpectralDoppler	Spectral Doppler with Audio (Pulsed Wave Doppler)					★L	★L							★L																					
WideBeam	Weakly Focused Overlapping Beams			★L	★L	★L	★L	★L	★	★		★	★	★	★L	★L	★L												★			★L	★L	★L	★L
WideBeamHI	Harmonic Imaging with Pulse Inversion			★L	★L	★L	★L	★L						★L	★L	★L															★L		★L	★L	
High Frequency Imaging										★																									
128RyLns67BW	Scanline Imaging — 67% BandWidth									★																									
Flash67BW	Single Plane Wave Imaging— 67% BandWidth									★																									
FlashAngles67BW	Multiple Plane Waves — 67% BandWidth																																		
128RyLnsInterleave	Scanline Imaging with 2x interleaved sampling																																		
FlashAnglesInterleave	Flash Angles with 2x interleaved sampling									★	★														★										
WideBeamInterleave	WideBeam with 2x interleaved sampling									★		★													★										
High Image Quality																																			
128RyLnsPolar	Conventional Focused Beams with Polar Coor.			★L	●										★L	●							●							★					
WideBeamHIPolar	Harmonic Imaging with Polar Coor.			★L								●	●		★L								●												
WideBeamPolar	Focused Overlapping Beams with Polar Coor.			★L	★L							●	●		★L	★L			★	●			●							★L	★L				
WideBeamPolarSC	WideBeam with Spatial Compounding with Polar Coor.				★											★																			
WideBeamDoppler	WideBeam Doppler with enhanced GUI controls			★	★L	★L	★L	★L	★	★		●	●	★L	★	★L		★	★	★		★			★							★L	★L	★L	★L

	System	Vantage 512		Vantage 256 / 128 / 64 LE																						Vantage 64 / 32 LE									
File Name (root)	Description of Example	UTA-512-Direct	UTA-2048-MUX	UTA-260-S and 260-D										UTA-408-GE										UTA-156-U		UTA-1024-MUX [2]		UTA-260-MUX							
Setup_Probe_root.m		L512tst	L512tst	P4-2v	C5-2v	L11-4v	L11-5v	L12-3v	L22-8v	L22-14v/vXL35-16v	L38-22v	IP-104	IP-105	L11-5gH	P4-2gH	C5-2gH	4CD	9LD	C1-6D	IC5-9D	L3-12D	M5ScDRC6gV	RC15gV	L39-21gD	L14-5u	SA4-2-24u	3 MHz		P4-2v	C5-2v	L11-4v	L11-5v	L22-14v/LF	[3]	
WideBeamHISC	WideBeamSC with Harmonic Imaging				★L	★L	★L	★L						★L		★L	★	★	★		★														
WideBeamSC	WideBeam with Spatial Compounding				★L	★L	★L	★L	★	★L				★L		★L	★	★	★		★											★L	★L	★L	★L
WideBeamSCInterleave	WideBeamSC with 2x interleaved sampling																								★										

● Not in the High Image Quality subfolder L: LE system

[1] L22-14v/vX also has a long focus version with one script available (Flash). Please refer to L22-14v/vX for other scripts.

[2] Special applications, sparse aperture and SRAC, are provided. All scripts can be easily modified for the 8-MHz 1024-MUX transducer

[3] L22-14 vX/LF scripts are also available for Vantage 64

[4] L39-21gD is only available for high frequency

[5] Example scripts found in Example_Scripts/Specialty_Applications folder.

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